# Kazem Faghih

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## Overview

I am a Ph.D. student in Computer Science at the University of Maryland, advised by Prof. Soheil Feizi. My research examines what benchmark scores miss about LLM reliability and how tool-using agents can be manipulated through tool and skill descriptions.

- **LLM Reliability & Evaluation:** Developing evaluations that distinguish models that answer correctly only under favorable wording from those that answer consistently—even when their benchmark scores are similar.
- **Agent Reliability & Security:** Studying how tool and skill descriptions influence agent decisions, from tool-selection bias to attacks on third-party skill registries.
- **Training & Systems:** Experience with continued pre-training and instruction tuning of language models, custom CUDA kernels for protein sequence search, and high-throughput production systems.

## Education

**University of Maryland, College Park** — College Park, MD
*Ph.D. in Computer Science; GPA: 4.00/4.00* — *Jan. 2025 – Expected Dec. 2028*
Advisor: Prof. Soheil Feizi

**Sharif University of Technology** — Tehran, Iran
*B.Sc. in Computer Engineering* — *2017 – 2022*

## Selected Publications

* denotes equal contribution.

Reliability & Evaluation

- **Revealing What Benchmark Scores Hide: Jointly Measuring the Breadth and Reliability of LLM Capabilities Using Accessibility Profiles**
  **K. Faghih**, Y. Cheng, S. Saha, M. Pournemat, A. Gerami, S. Feizi
  *Preprint, under review, 2026.* [link]

Reliable Agents & Tool Use

- **Tool Preferences in Agentic LLMs are Unreliable**
  **K. Faghih***, W. Wang*, Y. Cheng*, S. Bharti, G. Sriramanan, S. Balasubramanian, P. Hosseini, S. Feizi
  *EMNLP 2025.* [link]
- **Your LLM Agents are Temporally Blind: The Misalignment Between Tool Use Decisions and Human Time Perception**
  Y. Cheng*, A. S. Moakhar*, C. Fan*, P. Hosseini, **K. Faghih**, Z. Sodagar, W. Wang, S. Feizi
  *Findings of ACL 2026.* [link]

Agent Security

- **Under the Hood of SKILL.md: Semantic Supply-Chain Attacks on AI Agent Skill Registry**
  S. Saha, **K. Faghih**, S. Feizi
  *Preprint, under review, 2026.* [link]
  Media Coverage: The Register

Retrieval & Attribution

- **Quantifying Document Impact in RAG-LLMs**
  A. Gerami, **K. Faghih**, R. Duraiswami
  *Preprint, 2025.* [link]

## Honors & Service

**Dean's Fellowship**, University of Maryland — 2025–2026
**2nd / 145,000+**, Iranian Nationwide University Entrance Exam, Mathematics & Physics — 2017
**Reviewer:** ACL; NeurIPS.

## Research Experience

**University of Maryland** College Park, MD
*Research Assistant; Advisor: Prof. Soheil Feizi* *Jan. 2025 – Present*

- Initiated and led (first author) research on evaluating LLMs across rewordings of the same question, separating capabilities that are reliable from ones that appear only under favorable wording.
- Found that leaderboards can rank near-tied models the wrong way round: for about half of such pairs, which model is better depends on whether you need one correct answer or a consistent one.
- Co-led tool-selection research (co-first author, EMNLP 2025) showing that surface changes to tool descriptions can increase selection probability by more than 20× in some settings.

**Max Planck Institute for Multidisciplinary Sciences** Göttingen, Germany
*Research Assistant; Supervisor: Prof. Johannes Söding; with Prof. Martin Steinegger* *Aug. 2022 – Apr. 2023*

- Developed custom CUDA kernels for an ungapped local-alignment prefilter in MMseqs2 (used in AlphaFold and ColabFold workflows), mapping sequence-search computation onto GPU threads, blocks, and warps.
- Achieved a ∼2× speedup over the then-optimized MMseqs2 CPU implementation at comparable sensitivity.

**Sharif University of Technology** Tehran, Iran
*Research Assistant; Advisor: Prof. Ali Sharif-Zarchi* *Jun. 2022 – Jan. 2023*

- Implemented and evaluated peptide–HLA binding prediction using AlphaFold-derived representations; investigated protein language models and homology/MSA-based embeddings.

## Industry Experience

**Yektanet** Tehran, Iran
*Software Engineer* *Apr. 2024 – Nov. 2024*

- Built retention-marketing infrastructure supporting 100M+ contacts and 100K+ SMS/push messages per minute.
- Built a location service handling ∼4K requests/sec.

**Mofid Securities** Tehran, Iran
*ML Engineer & Data Scientist* *Jun. 2023 – Apr. 2024*

- Led adaptation of a generative LLM to Persian: data, tokenizer/architecture design, continued pre-training, instruction tuning, and evaluation.
- Solving classification generatively, reached 78 and 84 F1 on Persian sentiment tasks, within 4 points of ParsBERT (82 and 88), the leading task-specific encoder at the time.
- Built a RAG chatbot and ML-serving services.

**Cafebazaar / Balad Maps** Tehran, Iran
*Software Engineer & Data Engineer* *Jul. 2020 – Sep. 2021*

- Owned services processing ∼200M map records/day; migrated Django/REST services to Go/gRPC.

**Nimbo** Tehran, Iran
*Data Engineering & Big Data Intern* *Jun. 2019 – Sep. 2019*

- Built a distributed crawler/search prototype over ∼100M documents using Hadoop, Spark, Kafka, and Elasticsearch.

## Technical Skills

**ML / LLMs:** PyTorch, Hugging Face Transformers, vLLM, pre-training, supervised fine-tuning, LoRA/PEFT, prompt optimization, RAG, benchmark design.
**GPU / Systems:** CUDA kernels, SLURM, HPC, Docker, Kubernetes, Spark, Kafka, gRPC.
**Programming:** Python, C/C++, Go, SQL, Bash.